RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL

New Scheme Based On AICTE Flexible Curricula

Artificial Intelligence and Data Science, V-Semester

AD 504 (C) Operations Research

COURSE OUTCOMES: After Completing the course student should be able to:

CO1: Develop the concepts and able to formulate and solve Linear Programming Problems.
CO2: Design an optimal solution for transportation & Assignment Problem
CO3:Identify &analyze the scheduling of activities using PERT and CPM techniques.
CO4:Understand Job sequencing problems and the use of Dynamic programming in OR.
CO5: Develop a solution for queuing models.

COURSE CONTENTS:

UNIT-I

Origin & Development of OR, Different Phases of OR study, Methodology of OR, Scope andLimitations of OR, OR in decision making. Linear Programming: Introduction – Mathematical formulation of a problem –Graphical solutions, standard forms the simplex method for maximization andminimization problems, Interpretationof Duality, Dual simplex Method.

UNIT-II:

Transportation problem (TP) and its formulation. Finding basic feasible solution and optimal solution for transportation problem.

Assignment problem and its formulation, Hungarian methodfor solving Assignment problem, travelling salesmen problem.

UNIT-III

Project Scheduling: PERT and CPM with known activity times. Critical Path Analysis,Various types of floats. Probability considerations in PERT. Updating of PERT charts.Project crashing. Formulation of CPM as a linear programming problem. Resource levellingand resource scheduling.

UNIT-IV

Sequencing problem: Introduction to sequencing problem. Flow shop problem: Processing njobs through 2, 3 and m machines. General n/m job-shop problem.

Introduction to Dynamic Programming. Dynamic Programming Approach for solving Linear ProgrammingProblem. Applications of Dynamic programming.

UNIT-V

Queuing: Introduction to queuing theory, Queuing systems and their characteristics, Pure-birth and Pure-deathmodels, Kendall & Lee's notation of Queuing, empirical queuing models – Numerical on M/M/1 and M/M/C Queuing models.

TEXT BOOKS:

1. Operations Research, P K Gupta and D S Hira, S. Chand and Company LTD. Publications,New Delhi – 2007

2. Operations Research, An Introduction, Seventh Edition, Hamdy A. Taha, PHI PrivateLimited, 2006.

REFERENCE BOOKS:

1. Operations Research, Theory and Applications, Sixth Edition, J K Sharma, Trinity Press,Laxmi Publications Pvt. Ltd. 2016.

2. Operations Research, Paneerselvan, PHI

3. Operations Research, A M Natarajan, P Balasubramani, Pearson Education, 2005

4. Introduction to Operations Research, Hillier and Lieberman, 8th Ed., McGraw Hill